

SYMBOL		SERVICE	LINE SIZE
AIR SYSTEMS			
IA	INSTRUMENT AIR		
PA	PLANT AIR		
BLOWDOWN AND PUMPOUT SYSTEMS			
CABD	CAUSTIC BLOWDOWN		
CBD	COLD BLOWDOWN		
HBD	HOT BLOWDOWN		
PB	PUMPOUTS		
SB	SLOPS		
HOBD	HEAVY OIL BLOWDOWN		
STEAM AND CONDENSATE SYSTEMS			
DS	DILUTION STEAM		
ES	EXHAUST STEAM		
HS	HIGH PRESSURE STEAM		
HHS	HIGH HIGH PRESSURE STEAM		
LLS	LOW LOW PRESSURE STEAM		
LS	LOW PRESSURE STEAM		
MS	MEDIUM PRESSURE STEAM		
HC	HP CONDENSATE		
MC	MP CONDENSATE		
LC	LP CONDENSATE		
LLC	LOW-LOW PRESSURE CONDENSATE		
TC	TURBINE CONDENSATE		
TS	TRACING STEAM		
DRAIN SYSTEMS			
CD	CHEMICAL DRAIN		
DR	DRAIN		
LD	CRYOGENIC LIQUID DRAIN		
ND	NON-CONTAMINATED DRAIN (FROW CW, DEMIN WATER, NOT WASTE WATER TREATMENT REQUIRED)		
OD	OILY DRAIN / CASING DRAIN		
OWD	OILY WATER DRAIN		
SSB	SOLVENT SLOP DRAIN		
SD	STORMWATER DRAIN		
VENT SYSTEMS			
AV	ATMOSPHERIC VENT (NON PSV)		
SV	SAFETY VALVE TO ATMOSPHERE		
V	VENT		
FLARE SYSTEMS			
CFL	COLD FLARE GAS		
HFL	HOT FLARE GAS		
FL	FLARE		
GAS SYSTEMS			
DA	DECOKE AIR		
FG	FUEL GAS		
FL	FLUE GAS		
HN	HP NITROGEN		
NC	NONCONDENSABLE GAS		
NG	NATURAL GAS		
H	HYDROGEN GAS		
LN	LP NITROGEN (<= 0.8 MPag)		
OX	OXYGEN		
OG	OFFGAS		
SPECIAL LIQUID SYSTEMS			
AC	ACID		
CL	CAUSTIC		
CAS	CHEMICAL LIQUID		
CH	CHEMICAL		
CHL	CHILLED LIQUID		
NH	AMMONIA		
IL	CHEMICAL INJECTION (DEMULSIFYING, ETC.)		
ME	METHANOL		
OIL UTILITY SYSTEMS			
FO	FUEL OIL		
LO	LUBE OIL / SEAL OIL		
WO	WASH OIL / FLUSHING OIL		
QO	QUENCH OIL		
PROCESS SERVICE			
P	PROCESS (HYDROCARBONS)		
PC	PROCESS CONDENSATE		
PG	PROCESS GAS		
PL	PROCESS LIQUID		
PM	PROCESS (2 PHASE)		
PS	PROCESS SOLID		
REFRIGERANT SYSTEMS			
PEG	PROPYLENE GAS (FOR REFRIGERATION SYSTEM ONLY)		
PEL	PROPYLENE LIQUID (FOR REFRIGERATION SYSTEM ONLY)		
RP	PROPYLENE REFRIGERANT		

LINE SERVICE IDENTIFICATION

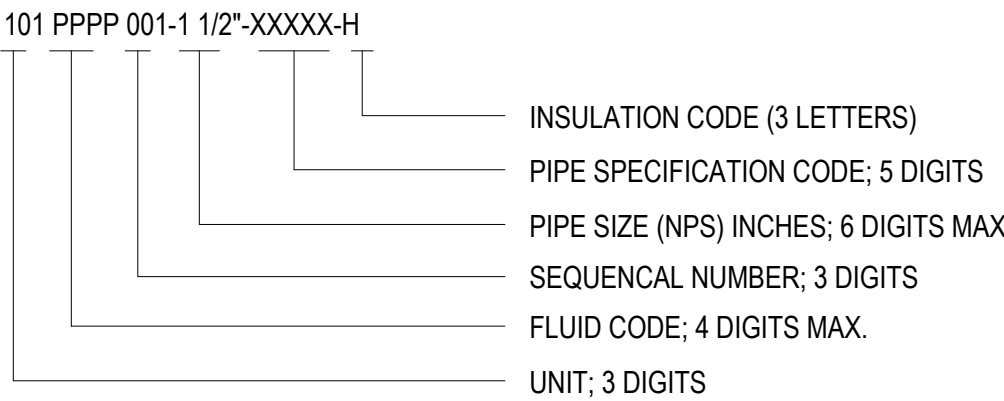
WATER SYSTEMS

BFW	BOILER FEED WATER	IW	INDUSTRIAL WATER
LP BFW	LP BOILER FEED WATER	PSW	STRIPPED WATER
MP BFW	MP BOILER FEED WATER	PW	POTABLE WATER
CWR	COOLING WATER RETURN	PWW	WASHED WATER
CWS	COOLING WATER SUPPLY	QW	QUENCH WATER
DW	DEMINERALIZED WATER	RW	CLEAN RAIN WATER
FW	FIRE FIGHTING WATER	SW	SOFTEN WATER
HWR	HOT WATER RETURN	OWW	OILY WASTE WATER
HWS	HOT WATER SUPPLY		

MISCELLANEOUS ABBREVIATIONS

A	AIR MOTOR	LC	LOCKED CLOSED
ACC	ACCESSIBLE	LO	LOCKED OPEN
AG	ABOVE GROUND	LP	LOW PRESSURE
ATM	ATMOSPHERE	LLP	LESS LOW PRESSURE
BL	BATTERY LIMITS	LLPD	DESUPERHEATED LESS LOW PRESSURE
BTS	BUBBLE TIGHT SHUT-OFF	M	ELECTRIC MOTOR
BV	BY VENDOR	MAX	MAXIMUM
CO	CLEAN OUT	MIN	MINIMUM
CONN	CONNECTION	MOR	MIDDLE OF RUN
CSC	CAR SEAL CLOSED	MP	MEDIUM PRESSURE
CSO	CAR SEAL OPEN	MPS2	INTERIM MEDIUM PRESSURE STEAM
C2R	ETHYLENE REFRIGERANT	MW	MANWAY
C3R	PROPYLENE REFRIGERANT	NC	NORMALLY CLOSED
CW	COOLING WATER (PFD ONLY)	NNF	NORMALLY NO FLOW
D	DRAIN	NO	NORMALLY OPEN
DC	DRAIN CONNECTION	NOZ	NOZZLE
DEC	DETAILED ENGINEERING CONTRACTOR	OSBL	OUTSIDE BATTERY LIMITS
DF	DRINKING FOUNTAIN	RO	ROD OUT
DNP	DO NOT POCKET	RSP	REMOVABLE SPOOL PIECE
EM	ELEVATED MONITOR	S	SAMPLE CONNECTION
EOR	END OF RUN	SC	SAMPLE CONNN WITH COOLER
EW	EYEWASH	SO	STEAM OUT
FLG	FLANGE	SOR	START OF RUN
FPRF	FIREPROOF	SS	SAFETY SHOWER
GO	GEAR OPERATED	T	LINE DRIP TRAP
H	HYDRAULIC	THD	THREADED
HC	HOSE CONNECTION	TL	TANGENT LINE
HH	HANDHOLE	T/T	TANGENT TO TANGENT
HP	HIGH PRESSURE	TS	TEMPORARY STRAINER
HR	HOSE REEL	TSO	TIGHT SHUTOFF
ID	INSIDE DIAMETER	TYP	TYPICAL
IF	INSULATING FLANGE	UC	UTILITY CONNECTION
INV	INVERT	UG	UNDERGROUND
ISBL	INSIDE BATTERY LIMITS	V	VENT
		VC	VENT CONNECTION

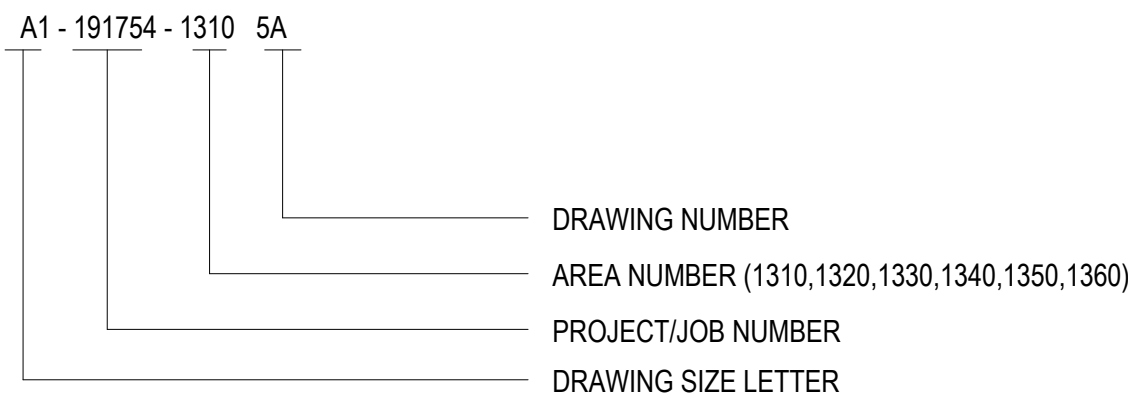
PIPELINE IDENTIFICATION



INSULATION CLASSIFICATION

A	-	ACOUSTIC
AH	-	ACOUSTIC INSULATION HOT
AC	-	ACOUSTIC INSULATION COLD
C	-	COLD INSULATION
E	-	ELECTRICAL TRACING
F	-	FULL HOT INSULATION
H	-	HOT INSULATION
M	-	DUAL TEMPERATURE
N	-	NO INSULATION
PP	-	PERSONNEL PROTECTION
PPC	-	PERSONNEL PROTECTION COLD
S4	-	LOW LOW PRESSURE STEAM TRACING
S10	-	LOW PRESSURE STEAM TRACING
W	-	HOT WATER TRACING
WN	-	WINTERIZATION

DRAWING NUMBER IDENTIFICATION



PIPING SYMBOLS

	PIPING LINES
	INSTRUMENTATION
	UNDERGROUND PIPING (BURIED LINE)
	TRACING
XX TRACING LEGEND	
EP	- ELECTRICAL TRACING FOR PROCESS REASONS (NOTE 4)
EW	- ELECTRICAL TRACING FOR WINTERIZATION
HW	- HOT WATER TRACING FOR WINTERIZATION
TL	- LP STEAM TRACING FOR PROCESS REASONS
TM	- MP STEAM TRACING FOR PROCESS REASONS
TW	- LP STEAM TRACING FOR WINTERIZATION
	FUTURE LINE
	JACKETED LINE
	VENDOR PACKAGE BATTERY LIMITS (OSBL)
	GATE VALVE
	VENTED GATE VALVE (VENT DIRECTION SHOWN)
	VALVE WITH BLIND OR PLUG
	BUTTERFLY VALVE
	DIAPHRAGM VALVE
	GLOBE VALVE
	NEEDLE VALVE
	PLUG VALVE
	BALL VALVE
	CHECK VALVE
	STOP CHECK VALVE (BOILER NON-RETURN VALVE)
	THREE WAY VALVE
	FOUR WAY VALVE
	MULTI PORT PLUG
	ANGLE VALVE
	DUAL TANDDEM BLOWDOWN VALVE
	ANGLE VALVE
	RAM TYPE VALVE (SPECIALTY ITEM)
	INTERLOCKED VALVES (MECHANICAL LINK)
	SPRING LOADED QUICK OPENING OR CLOSING VALVE
	SLIDE VALVE
	Y-TYPE GLOBE VALVE
	AUTOMATIC RECIRCULATION VALVE
	QUICK OPENING OR CLOSING VALVE
	VALVE WITH BODY CAVITY RELIEF OR BODY BLEED CONNECTION

	OSBL/ISBL
	CONTINUITY ARROW
	BLIND FLANGE
	PIPE CAP (WELDED)
	PIPE CAP (SCREWED)
	REDUCTION IN LINE SIZE
	TEMPORARY STRAINER
	CONE TYPE STRAINER
	T-TYPE STRAINER
	Y-TYPE STRAINER
	REMOVABLE SPOOL
	SPACER FOR BLIND
	LINE BLIND (SPADE)
	REVERSIBLE BLIND (OPEN)
	REVERSIBLE BLIND (CLOSED)
	SPEC BREAK XXX - PIPING CLASS
	MINIMUM DISTANCE
	DUCT OR STACK DAMPER
	OPEN SERVICE DRAIN XX - SERVICE SYMBOL
	CLOSED SERVICE DRAIN XX - SERVICE SYMBOL
	ATMOSPHERIC VENT
	FLAME ARRESTOR
	GRADE
	TRENCH
	SLOPE INDICATION
	PULSATION DAMPENR
	CHILLING TRAIN FILTER
	FLANGED CONNECTION
	HOSE CONNECTION
	ROD OUT CONNECTION
	CLEAN OUT CONNECTION (FLUSHING)
	ORIFICE SWAGE
	INSULATED FLANGED JOINT
	FLEXIBLE HOSE
	EXPANSION JOINT
	INLINE MIXER
	MIXING "T"
	HYDROMETER POT
	VORTEX BREAKER
	EXHAUST HEAD
	STEAM TRAP (NON PROCESS)
	PROCESS STEAM TRAP
	EDUCTOR OR EJECTOR
	PIPING SPECIALTY ITEM
	SEQUENCE NUMBER
	TIE-POINT
	SAFETY SHOWER
	UTILITY STATION
	SAMPLE CONN WITH COOLER
	TYPE
	SEQUENCE NUMBER
	LINE FILTER
	SILENCER
	LOW DB NOISE CARTRIDGE
	AIR FILTER WITH HOOD
	DESUPERHEATER
	FIRE HYDRANT
	FIRE HYDRANT WITH MONITOR
	HOSE HOUSE
	HOSE REEL

NOTES:

1. FOR LEGENDS, SYMBOLS, GENERAL NOTES AND ENGINEERING DESIGN NOTES SEE DWG. No. A1-191754-00005A, 00005B, 00005C AND 00005D.
2. ALL INSTRUMENT TAG NUMBERS WILL BE PRECEDED BY 4-DIGIT UNIT NUMBER UNLESS OTHERWISE SPECIFIED.
3. ALL DIMENSIONS ARE IN mm, WHILE ALL NOZZLE SIZES, LINE SIZES ARE IN inch, UNLESS OTHERWISE SPECIFIED.
4. ELECTRIC TRACING FOR PROCESS LINES SHOULD BE SELF LIMITING TO PREVENT OVERHEATING.